

Brianna Wimer

EDUCATION

University of Notre Dame, Notre Dame, IN
Ph.D. in Computer Science and Engineering
Advised by Dr. Ronald Metoyer

August 2021 - PRESENT

The University of Alabama, Tuscaloosa, AL
B.S in Computer Science

July 2018 - July 2021

INTERESTS

My research lies at the intersection of Accessibility and Visualization, with a primary focus on improving how individuals with disabilities interact with visualizations and diagrams. I design and develop systems that transform diagrams into accessible representations. My work aims to create tools that not only enhance data accessibility but also empower individuals with disabilities to actively create, interpret, and engage with visual diagrams.

RESEARCH EXPERIENCES

Visiting Student Researcher, The University of Washington
Advisor: Dr. Jennifer Mankoff

Fall 2023 - PRESENT

Graduate Research Assistant, The University of Notre Dame
Advisor: Dr. Ronald Metoyer

Fall 2021 - PRESENT

CRA-WP Distributed Research for Undergraduates (DREU)
Research Intern, University of Notre Dame; Advised by Dr. Ronald Metoyer

Summer 2021

CRA-WP Distributed Research for Undergraduates (DREU)
Research Intern, Tufts University; Advised by Dr. Elaine Short

Summer 2020

Undergraduate Research Assistant, The University of Alabama
Advisor: Dr. Chris Crawford

2019 to 2021

PUBLICATIONS

Glazko, K., Cha, J., Lewis, A., Kosa, B., **Wimer, B.**, Zheng, A., Zheng, R., Mankoff, J.
"Autoethnographic Insights from Neurodivergent GAI "Power Users"." In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pp. 1-19. 2025.

Wimer, B., Mack, KA., Metoyer, R., and Mankoff, J. "Accessible Flowcharts: A Feasibility Study with BVI Participants." *The Journal on Technology and Persons with Disabilities* (2025): 317.

Wimer, B., Szymanski, A., Metoyer, R. "Beyond Static Labels: Unpacking Nutrition Comprehension in the Digital Age." *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 2024

Szymanski, A., **Wimer, B.**, Anuyah, O., Eicher-Miller, H. A., & Metoyer, R. A. "Integrating Expertise in LLMs: Crafting a Customized Nutrition Assistant with Refined Template Instructions." *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 2024.

Ning, Z., **Wimer, B.**, Jiang, K., Chen, K., Ban, J., Tian, Y., Zhao, Y., Li, T. "SPICA: Interactive Video Content Exploration through Augmented Audio Descriptions for Blind or Low-Vision Viewers." *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 2024.

Wimer, B., South, L, K. Wu, D. A. Szafir, M. A. Borkin and R. Metoyer, "Beyond Vision Impairments: Redefining the Scope of Accessible Data Representations." *IEEE Transactions on Visualization and Computer Graphics* (2024).

Dillahunt, T., Sawwan, M., Wood, D., **Wimer, B.**, Conrado, A., Eicher-Miller, H., Zornig Gora, A., Metoyer, R. "Understanding Food Planning Strategies of Food Insecure Populations: Implications for Food-Agentic Technologies." *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 2023.

POSTER PRESENTATIONS

Wimer, B. *Understanding How Information Visualization Can Support Diverse Cognitive Abilities*. Poster Presented At: ACM Conference on Computers and Accessibility. 2022 Oct 24-26. Athens, Greece. (Not peer-reviewed, poster presentation for SIGACCESS Travel Scholarship).

Wimer, B., Szymanski, A. Food Information Networks (FINs): The Visual Representation of Food Information for Healthy Dietary Choices. Poster Presented At: Lucy Institute Fall Symposium. 2021 Oct 27. University of Notre Dame.

Wimer, B. *Improving Self-Efficacy in Emotion Regulation Through a Biofeedback Video Game*. Poster Presented At: Virtual Grace Hopper Celebration. 2020 Sept 29-Oct 3. Remote

AWARDS & ACHIEVEMENTS

Graduate Fellowship Award, Washington NASA Space Grant Consortium | 2024

Google Ph.D. Fellowship, Google | 2023

IEEE VIS Inclusivity & Diversity Scholarship, IEEE VIS | 2023

Graduate Cohort for IDEALS Travel Grant, Computing Research Association | 2023
SIGACCESS Travel Scholarship, ACM SIGACCESS | 2022
Graduate Cohort for IDEALS Travel Grant, Computing Research Association | 2022
Jack and Mary Ann Remick Fellowship, University of Notre Dame | 2021
Louis Stokes Alliance for Minority Participation Scholar, University of Alabama | 2018 -2021
Dean's List, The University of Alabama | 2021, 2020
Tapia Travel Grant, AccessComputing | 2020
Grace Hopper Travel Grant, AccessComputing | 2020

TEACHING EXPERIENCE

University of Notre Dame

2023 **Warrior-Scholar Project Camp For Veterans and Active Duty Service Members**
Role: Instructor / Research Project Leader

2022 **Fundamentals of Computing (CSE20311)**
61 Undergraduate students enrolled. Role: Graduate Teaching Assistant

2021 **Academic Services for Student Athletes**
2 Undergraduate students assigned to me. Role: Strategy Tutor

2021 **Academic Services for Student Athletes**
12 Undergraduate students assigned to me. Role: Calculus Tutor

INVITED TALKS

2024 **Guest Speaker**, Accessibility Lunch at Carnegie Mellon University

2023 **Panelist**, Inclusive Design x The Alliance for Identity-Inclusive Computing Education

2021 **Panelist**, Code.Org x CareerVillage
Panelist, Explore STEM@UTSA

2020 **Guest Speaker**, Louis Stokes Alliance for Minority Participation
Guest Speaker, Bridge to Engineering Success at Tufts Webinar

ACADEMIC SERVICE

Professional Service

2025 **Lead Organizer**: Accessible Visualization Workshop at IEEE VIS'25
2024 **Publication Reviewer**: UIST, DIS
2022 **Student Volunteer**: CHI, DIS, ASSETS

University and Community Service

2024 **Graduate Leader**, UW NASA Space Grant Consortium

2023	Summer Program Co-Director , ND CSE Summer Enrichment Program Counselor/Mentor , UA's LEGACY Project
2022	Graduate Research Mentor , ND's Center for Civic Innovation Interns Graduate Research Mentor , ND's iTREDS Program
2020	Research Mentor , UA's Louis Stokes Alliance for Minority Participation
2019	Counselor/Mentor , UA's LEGACY Project Counselor , UA's NeuroCamp

NEWS MEDIA COVERAGE

2023	ND Engineering News. <u>"Brianna Lynn Wimer named 2023 Google Ph.D. Fellow to address computing needs of vision-impaired people"</u>
2021	AccessComputing STEM For All Video Showcase.
2020	AccessComputing Newsletter Profile. <u>"Distributed Research Experience for Undergraduates (DREU): Firsthand Benefits"</u>
2019	UA Engineering News. <u>"LEGACY program prepares young women of color with computer science education"</u>